

Tanvir Ahmed

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Skills

Languages: Python, C, C++, Java, JavaScript, Dart, PHP, SQL, PostgreSQL

AI / ML & Libraries: TensorFlow, PyTorch, scikit-learn, Keras, OpenCV, Mediapipe, NumPy, Pandas, Selenium, Tkinter, Matplotlib, Seaborn, LangChain, ChromaDB, Qdrant.

Web / Mobile / Tools: HTML, CSS, Bootstrap, Tailwind CSS, JavaScript, Flutter, Firebase, PHP, Git, GitHub, VS Code, Figma, OpenGL Google Colab, Kaggle, Tinkercad, Zapier, Make.

Concepts & Domains: Machine Learning, Deep Learning, Computer Vision, RAG, Image/Video Processing, NLP, Object Tracking, IoT, Automation, Human-Computer Interaction, Data Visualization, Big Data, Information Retrieval.

Work Experience

AI/ML Intern - Digital Technology & Innovation (DTI)

2026 (February)–2026 (May)

United Commercial Bank PLC.

- Built a production-grade multilingual RAG chatbot (English/Bangla/Banglish) for customer support using hybrid retrieval (dense embeddings + BM25 + reranking) and a fully local pipeline (scraping → preprocessing → embedding → vector indexing); implemented context control and query filtering to reduce hallucination and improve response accuracy.
- Developed a hybrid document understanding pipeline combining OCR, layout-aware region segmentation, template matching, and human-in-the-loop validation to improve handwritten text extraction accuracy, minimize manual processing overhead, and streamline banking workflow automation at scale.
- Developed a fraud detection system for real-time transaction monitoring with risk scoring based on velocity, frequency, and behavioral signals; designed a scalable architecture for future ML-based anomaly detection and customer behavioral profiling.

Student / Researcher (Freelance & Academic Projects)

2024–2025

Independent Projects & Research

- Built and implemented applied ML and automation projects during B.Sc.; collaborated with supervisors on data set curation, model design, and end-to-end apps.
- Designed and implemented diverse academic, personal, and client projects, applying practical ML and software engineering skills to real-world problems.

Education

Daffodil International University

2022–2025

B.Sc. in Computer Science & Engineering

Relevant Coursework: Object-Oriented Programming, Databases, Discrete Mathematics, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning and Data Mining, Image Processing, Computer Graphics, System Design, Software Engineering, Information Security, Statistics and Probability.

Cantonment College Jashore

2018–2020

Higher Secondary Certificate (HSC)

GPA: 5.00

Nabojibon Institute Satkhira

2016–2018

Secondary School Certificate (SSC)

GPA: 5.00

Project Work

CrimeMap BD

2024–2026

Python, Flask, PostgreSQL, JavaScript, Leaflet.js, Vercel, Supabase

- Developed a bilingual, responsive dashboard with geospatial mapping (Leaflet.js), dynamic filtering, and automated classification of 15+ Bangladeshi news sources across 64 districts using parallel HTTP fetching (32x speed improvement) and NLP-based deduplication
- Built interactive geospatial dashboard with Leaflet.js clustering, heatmap, and Chart.js analytics; deployed serverless on Vercel + Supabase with sub-100ms API response times
- Maintained high code reliability by implementing a robust test suite covering 91+ API routes, text classification pipelines, and clustering logic.

Automation Tool for Levi Strauss & Co.

2024–2025

Python, Selenium, Tkinter

- Interactive app to upload Excel and auto-fill web forms with dynamic field mapping, reducing manual entry and errors.

AgroKart BD: Cross-Platform Marketplace2024–2025Flutter, Dart, Firebase, HTML/CSS/JS, PHP, SQL

- Developed a mobile app (Flutter) and a responsive web portal connecting farmers directly with consumers.
- Implemented user authentication, product listings, cart and backend order management (PHP & SQL).

Smart Home Automation System: Mobile App and Voice Controlled2024–2025ESP32, Arduino (C++), Flame Sensor, MQ-5, DHT11, LDR

- Built a smart home IoT automation system enabling remote control of lights, TV, fan, and AC via mobile app and voice assistants (Google Assistant and Amazon Alexa).
- Implemented real-time sensor monitoring (temperature, humidity) with dynamic UI updates.
- Developed timer and scheduler features supporting auto-shutdown (e.g., power-off after 20 minutes) and custom daily routines. Also allow users to create automated behaviors and configurable triggers.

Edu-Buddy: AI Student Assistant2023–2024Python, SpeechRecognition, Google TTS, SQLite, Tkinter

- Voice-activated assistant for reminders, study schedules, and quick access to educational resources.
- Modular intent parsing and dialog flow; integrated TTS and speech recognition for desktop use.

Hand Gesture Virtual Mouse2023–2024Python, OpenCV, Mediapipe, PyAutoGUI

- Real-time gesture-controlled cursor with click, scroll, and drag using multi-landmark detection.
- Optimized frame pipeline for low-latency interaction on consumer hardware.

Cross-Platform Task Manager2023–2024Python, psutil, Tkinter

- Lightweight tool to monitor processes, CPU/memory usage and perform safe terminations across platforms.
- Optimized system resource utilization, yielding a 10–12% reduction in CPU/memory overhead across platforms.

Research & Publications

Towards Automatic Classification and Translation of Bangla Regional Dialects in Low-Resource Settings2024–Present

Natural Language Processing, Deep Learning, Transformer Models

- Built and curated one of the largest Bangla regional dialect datasets containing 20,090 unique samples across 8 dialect classes.
- Developed a deep learning-based NLP system for automatic Bangla dialect classification and translation in low-resource settings.
- Achieved 86% multi-class classification accuracy using transformer-based and deep learning architectures.

Truth in Pixel: Deepfake Detection2024–Present

Computer Vision, Deep Learning

- Working on pixel-level inconsistency detection and robust artifact-based pipelines for manipulation detection.

Exploiting Transformer Architectures for Domain-Specific Text Categorization: The m-BERT Perspective2024–2025

Transformer Learning, Deep Learning, Natural Language Processing

- Achieved benchmarking accuracy on this research among all existing works.

Plant Leaf Disease Recognition Dataset2023–2024

Mendeley Data: Plant Leaf Disease Recognition Dataset

- Curated and published 20000+ annotated plant leaf images (zucchini, hibiscus, gourd, papaya) to support CV research.

Deep Learning Models for Multi-Class Plant Leaf Disease Classification2023–2024

CNN, Image Classification

- Trained Multiple CNNs achieving validation accuracy exceeding 97% using augmentation and tuning.

Awards and Certificates

Selected Achievements

- First Position — District Level Science Fair (Innovation Category)
- Third Position — Division Level Science Fair (Innovation Category)
- 9th Position — Rotaract science fair at Jashore University of Science and Technology (college-level).